AP/ITEC 4220 A - Modern Approaches to Data Management: Database Management Systems

Milestone 2

Group 2

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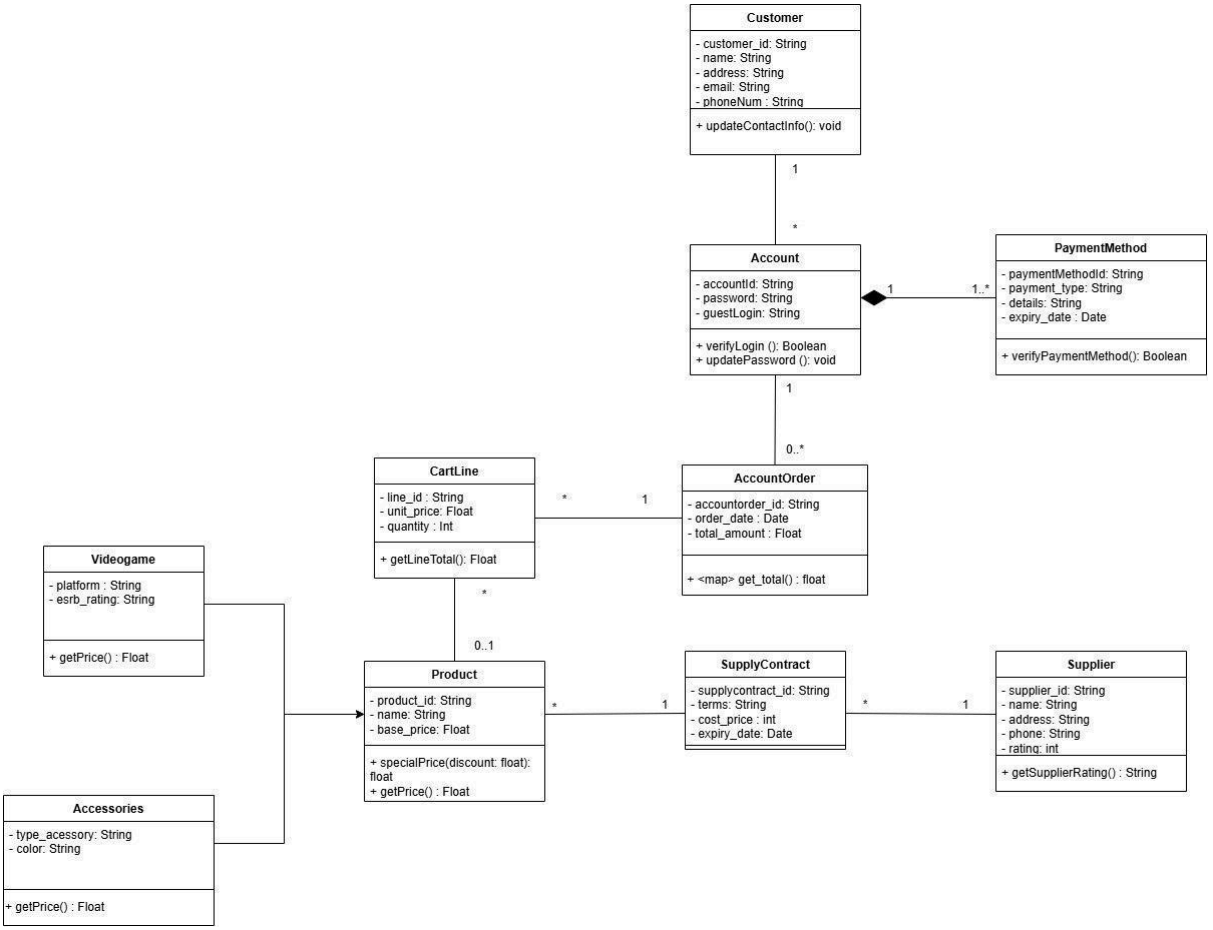
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Revised Class Diagram



Creating Tables:

Create object tables (of the types)

Create table customer of customer_t (primary key (customer_id));

Create table account of account_t (primary key (account_id));

Create table paymentmethod of paymentmethod_t (primary key (paymentmethod_id));

Create table videogame of videogame_t (primary key (product_id));

Create table accessories of accessories_t (primary key (product_id));

Create table supplycontract of supplycontract_t (primary key (supplycontract_id));

Create table accountorder of accountorder_t (primary key (accountorder_id));

Create table cartline of cartline_t (primary key (cartline_id));

Create table supplier of supplier_t (primary key (supplier_id));

Customer Type

```
CREATE TYPE customer_t AS OBJECT (
    customer_id CHAR(5),
    name VARCHAR2(30),
    address VARCHAR2(100),
    email VARCHAR2(100),
    phoneNum CHAR(10),
    MEMBER PROCEDURE UpdateContactInfo(p_address VARCHAR2, p_email
VARCHAR2, p_phoneNum CHAR)
);
/
```

Account Type

```
CREATE TYPE account_t AS OBJECT (
    account_id CHAR(5),
    password VARCHAR2(20),
    guestLogin CHAR(1),
    customer_ref REF customer_t,
    MEMBER FUNCTION VerifyLogin(p_accountId VARCHAR2, p_password
VARCHAR2) RETURN CHAR,
    MEMBER PROCEDURE updatePassword(p_newPassword VARCHAR2)
```

);

/

Payment Method Type

```
CREATE TYPE paymentmethod_t AS OBJECT (
   paymentmethod_id CHAR(8),
   payment_type CHAR(10),
   details VARCHAR2(60),
   expiry_date DATE,
   MEMBER FUNCTION verifyPaymentMethod RETURN CHAR
);
/
```

Supplier Type

```
CREATE TYPE supplier_t AS OBJECT (
   supplier_id CHAR(4),
   name VARCHAR2(30),
   address VARCHAR2(100),
   phone CHAR(10),
   rating CHAR(2),
   MEMBER FUNCTION GetSupplierRating RETURN CHAR
);
/
```

Supply Contract Type with ORDER Method

```
CREATE TYPE supplycontract_t AS OBJECT (
   supplycontract_id CHAR(4),
   terms VARCHAR2(50),
   cost_price NUMBER,
```

```
    expiry_date DATE,
    supplier_ref REF supplier_t,
ORDER MEMBER FUNCTION compare_supplier(other_contract IN supplycontract_t)
RETURN INTEGER
);
/
```

Product Supertype

```
CREATE TYPE product_t AS OBJECT (
    product_id CHAR(5),
    name VARCHAR2(20),
    base_price NUMBER,
    supplycontract_ref REF supplycontract_t,
    MEMBER FUNCTION GetPrice RETURN NUMBER,
    MEMBER FUNCTION SpecialPrice(p_discount NUMBER) RETURN NUMBER
) NOT FINAL;
/
```

Videogame Subtype

```
CREATE TYPE videogame_t UNDER product_t (
    platform VARCHAR2(20),
    estb_rating CHAR(2),
    OVERRIDING MEMBER FUNCTION GetPrice RETURN NUMBER
);
/
```

Accessories Subtype

```
CREATE TYPE accessories_t UNDER product_t (
    accessory_type VARCHAR2(20),
```

```
    color VARCHAR2(10),
    OVERRIDING MEMBER FUNCTION GetPrice RETURN NUMBER
);
/
```

Account Order Type with REFs and MAP method

```
CREATE TYPE accountorder_t AS OBJECT (
    accountorder_id CHAR(6),
    order_date DATE,
    total_amount NUMBER,
    account_ref REF account_t,
    MAP MEMBER FUNCTION get_total RETURN NUMBER
);
/
```

Cart Line Type with REFs

```
CREATE TYPE cartline_t AS OBJECT (
    cartline_id CHAR(7),
    order_ref REF accountorder_t,
    product_ref REF product_t,
    quantity INTEGER,
    unit_price NUMBER,
    MEMBER FUNCTION GetLineTotal RETURN NUMBER
);
/
```

TYPE Bodies:

Customer type body

```
CREATE TYPE BODY customer_t AS MEMBER PROCEDURE
UpdateContactInfo(p_address VARCHAR2, p_email VARCHAR2, p_phoneNum CHAR) IS
   BEGIN
      IF p_address IS NOT NULL THEN
         self.address := p_address;
      END IF;

      IF p_email IS NOT NULL THEN
         self.email := p_email;
      END IF;

      IF p_phoneNum IS NOT NULL THEN
         self.phoneNum := p_phoneNum;
      END IF;
   END UpdateContactInfo;
END;
/
```

Account Type Body

```
CREATE TYPE BODY account_t AS

   MEMBER FUNCTION VerifyLogin(p_accountId VARCHAR2, p_password

VARCHAR2) RETURN CHAR IS

   BEGIN

      IF self.account_id = p_accountId AND self.password = p_password THEN

         RETURN 'Y';

      ELSE

         RETURN 'N';

      END IF;

   END;

   MEMBER PROCEDURE updatePassword(p_newPassword VARCHAR2) IS

   BEGIN
```

```
        self.password := p_newPassword;
        UPDATE account a
        SET a.password = self.password
        WHERE a.account_id = self.account_id;
    END;
END;
/
```

Payment Method Type Body

```
CREATE TYPE BODY paymentmethod_t AS
    MEMBER FUNCTION verifyPaymentMethod RETURN CHAR IS
    BEGIN
        IF self.expiry_date > SYSDATE THEN
            RETURN 'Y';
        ELSE
            RETURN 'N';
        END IF;
    END;
END;
/
```

Product Type Body

```
CREATE TYPE BODY product_t AS
    MEMBER FUNCTION GetPrice RETURN NUMBER IS
    BEGIN
        RETURN self.base_price;
    END;
```

```
    MEMBER FUNCTION SpecialPrice(p_discount NUMBER) RETURN NUMBER IS
    BEGIN
        RETURN self.base_price * (1 - p_discount/100);
    END;
END;
/
```

Videogame Type Body (Override Method)

```
CREATE TYPE BODY videogame_t AS
    OVERRIDING MEMBER FUNCTION GetPrice RETURN NUMBER IS
        v_price NUMBER;
    BEGIN
        v_price := self.base_price;
        IF LOWER(self.platform) = 'steam' THEN
            v_price := v_price * 0.9;
        END IF;
        RETURN v_price;
    END;
END;
/
```

Accessories Type Body (Override Method)

```
CREATE TYPE BODY accessories_t AS
    OVERRIDING MEMBER FUNCTION GetPrice RETURN NUMBER IS
        v_price NUMBER;
    BEGIN
        v_price := self.base_price;
        IF LOWER(self.color) = 'gold' THEN
            v_price := v_price + 10;
```

```
    END IF;
    RETURN v_price;
  END;
END;
/
```

Supplier Type Body

```
CREATE TYPE BODY supplier_t AS
  MEMBER FUNCTION GetSupplierRating RETURN CHAR IS
  BEGIN
    RETURN self.rating;
  END;
END;
/
```

Supply Contract Type Body with Order Method

```
CREATE TYPE BODY supplycontract_t AS
  ORDER MEMBER FUNCTION compare_supplier(other_contract IN supplycontract_t)
  RETURN INTEGER IS
    result INTEGER := 0;
  BEGIN
    IF self.cost_price < other_contract.cost_price THEN
      result := -1;
    ELSIF self.cost_price > other_contract.cost_price THEN
      result := 1;
    END IF;
    RETURN result;
  END compare_supplier;
END;
/
```

Account Order Type Body

```
CREATE TYPE BODY accountorder_t AS
  MAP MEMBER FUNCTION get_total RETURN NUMBER IS
```

```
    BEGIN
        RETURN self.total_amount;
    END;
END;
/
```

Cartline Type Body

```
CREATE TYPE BODY cartline_t AS
    MEMBER FUNCTION GetLineTotal RETURN NUMBER IS
    BEGIN
        RETURN self.unit_price * self.quantity;
    END;
END;
/
```

Insert Data (3-4 for each object table)

Customer

```
Insert into customer values (customer_t('00001', 'Johnson Smith', '123 Queen St W',
'johnsons92@hotmail.com', '8475928528'));
Insert into customer values (customer_t('05011', 'Michael Nguyen', '940 King St W',
'michaelnguyen@gmail.com', '4378492957'));
Insert into customer values (customer_t('00325', 'Lucas Thompson', '55 Front St E',
'lucaslucas@gmail.com', '6472859274'));
Insert into customer values (customer_t('00123', 'Keanu Reeves', '156 Hollywood Dr',
'kreeves@hotmail.com', '4485163413'));
```

INSERT INTO customer VALUES (customer_t('00326', 'Sarah Mitchell', '91 Yonge St', 'sarah.mitchell@outlook.com', '4165551234'));

Account with REFs

INSERT INTO account VALUES (account_t('A0001', 'password123', 'N', (SELECT REF(c) FROM customer c WHERE c.customer_id = '00001')));

INSERT INTO account VALUES (account_t('A0002', 'secure456', 'N', (SELECT REF(c) FROM customer c WHERE c.customer_id = '00123')));

INSERT INTO account VALUES (account_t('A0003', 'guestpass', 'Y', (SELECT REF(c) FROM customer c WHERE c.customer_id = '00325')));

INSERT INTO account VALUES (account_t('A0004', 'matrix123', 'N', (SELECT REF(c) FROM customer c WHERE c.customer_id = '05011')));

INSERT INTO account VALUES (account_t('A0005', 'york789', 'N', (SELECT REF(c) FROM customer c WHERE c.customer_id = '00326')));

Payment Method

INSERT INTO paymentmethod VALUES (paymentmethod_t('00008462', 'e-transfer', 'J48NSFK83HU3', LAST_DAY(TO_DATE('01/04/2026', 'DD/MM/YYYY'))));

INSERT INTO paymentmethod VALUES (paymentmethod_t('00084729', 'card', 'VISA', LAST_DAY(TO_DATE('30/12/2026', 'DD/MM/YYYY'))));

INSERT INTO paymentmethod VALUES (paymentmethod_t('00445894', 'e-transfer', 'H94D73KDSNF8', LAST_DAY(TO_DATE('27/09/2026', 'DD/MM/YYYY'))));

Suppliers

INSERT INTO supplier VALUES (supplier_t('S001', 'GameDistributors Inc', '789 Supply Ave', '4165551234', 'A'));

INSERT INTO supplier VALUES (supplier_t('S002', 'TechParts Ltd', '456 Industry St', '4165555678', 'B'));

INSERT INTO supplier VALUES (supplier_t('S003', 'Accessory World', '321 Commerce Rd', '4165559012', 'A'));

INSERT INTO supplier VALUES (supplier_t('S004', 'Digital Games Co', '654 Online Blvd', '4165553456', 'C'));

Supply Contract with REFs

Insert into supplycontract values (supplycontract_t('S010', 'only gaming accessories', 25, DATE '2026-03-15', (SELECT REF(s) FROM supplier s WHERE s.supplier_id = 'S003')));

Insert into supplycontract values (supplycontract_t('S026', 'exclusive game distributor', 50, DATE '2025-12-01', (SELECT REF(s) FROM supplier s WHERE s.supplier_id = 'S001')));

INSERT INTO supplycontract VALUES (supplycontract_t('S029', 'send report every three months', 40, DATE '2028-08-21', (SELECT REF(s) FROM supplier s WHERE s.supplier_id = 'S004')));

Insert into supplycontract values (supplycontract_t('S018', 'ensure 98% on time delivery', 15, DATE '2026-08-15', (SELECT REF(s) FROM supplier s WHERE s.supplier_id = 'S002')));

Videogame with REFs

INSERT INTO videogame VALUES (videogame_t('G0001', 'Cyberpunk 2077', 59.99, (SELECT REF(s) FROM supplycontract s WHERE s.supplycontract_id = 'S026'), 'PS5', 'M'));

INSERT INTO videogame VALUES (videogame_t('G0002', 'Mario Kart', 49.99, (SELECT REF(s) FROM supplycontract s WHERE s.supplycontract_id = 'S026'), 'Switch', 'E'));

INSERT INTO videogame VALUES (videogame_t('G0003', 'Flight Simulator', 69.99, (SELECT REF(s) FROM supplycontract s WHERE s.supplycontract_id = 'S029'), 'Steam', 'E'));

INSERT INTO videogame VALUES (videogame_t('G0004', 'Call of Duty', 69.99, (SELECT REF(s) FROM supplycontract s WHERE s.supplycontract_id = 'S026'), 'XBOX', 'M'));

Accessories with REFs

INSERT INTO accessories VALUES (accessories_t('A0001', 'HyperX Cloud', 129.99, (SELECT REF(s) FROM supplycontract s WHERE s.supplycontract_id = 'S010'), 'headset', 'black'));

INSERT INTO accessories VALUES (accessories_t('A0002', 'Charge Type C', 29.99, (SELECT REF(s) FROM supplycontract s WHERE s.supplycontract_id = 'S018'), 'cable', 'white'));

INSERT INTO accessories VALUES (accessories_t('A0003', 'PS5 Controller', 79.99, (SELECT REF(s) FROM supplycontract s WHERE s.supplycontract_id = 'S010'), 'gamepad', 'gold'));

INSERT INTO accessories VALUES (accessories_t('A0004', 'Logitech GPX2', 59.99, (SELECT REF(s) FROM supplycontract s WHERE s.supplycontract_id = 'S010'), 'mouse', 'silver'));

Account Orders with REFs

INSERT INTO accountorder VALUES (accountorder_t('O00001', DATE '2025-10-15', 179.97, (SELECT REF(a) FROM account a WHERE a.account_id = 'A0001')));

```
INSERT INTO accountorder VALUES (accountorder_t('O00002', DATE '2025-10-18', 69.99,
   (SELECT REF(a) FROM account a WHERE a.account_id = 'A0004')));
INSERT INTO accountorder VALUES (accountorder_t('O00003', DATE '2025-10-20',
129.99, (SELECT REF(a) FROM account a WHERE a.account_id = 'A0002')));
INSERT INTO accountorder VALUES (accountorder_t('O00004', DATE '2025-10-22',
189.98, (SELECT REF(a) FROM account a WHERE a.account_id = 'A0003')));
```

CartLine with REFs

```
INSERT INTO cartline VALUES (cartline_t('CL00001',
   (SELECT REF(o) FROM accountorder o WHERE o.accountorder_id = 'O00001'),
   (SELECT REF(v) FROM videogame v WHERE v.product_id = 'G0001'),
   1, 59.99));
INSERT INTO cartline VALUES (cartline_t('CL00002',
   (SELECT REF(o) FROM accountorder o WHERE o.accountorder_id = 'O00001'),
   (SELECT REF(a) FROM accessories a WHERE a.product_id = 'A0001'),
   1, 129.99));
INSERT INTO cartline VALUES (cartline_t('CL00003',
   (SELECT REF(o) FROM accountorder o WHERE o.accountorder_id = 'O00002'),
   (SELECT REF(v) FROM videogame v WHERE v.product_id = 'G0003'),
   1, 69.99));
INSERT INTO cartline VALUES (cartline_t('CL00004',
   (SELECT REF(o) FROM accountorder o WHERE o.accountorder_id = 'O00003'),
   (SELECT REF(v) FROM videogame v WHERE v.product_id = 'G0004'),
   1, 69.99));
INSERT INTO cartline VALUES (cartline_t('CL00005',
```

```
    (SELECT REF(o) FROM accountorder o WHERE o.accountorder_id = 'O00004'),
    (SELECT REF(a) FROM accessories a WHERE a.product_id = 'A0003'),
    1, 79.99));
INSERT INTO cartline VALUES (cartline_t('CL00006',
    (SELECT REF(o) FROM accountorder o WHERE o.accountorder_id = 'O00004'),
    (SELECT REF(a) FROM accessories a WHERE a.product_id = 'A0003'),
    1, 79.99));
```

Query

Q1: Videogame platform ordered by customer Keanu Reeves

```
SELECT TREAT(DEREF(c.product_ref) AS videogame_t).platform AS platform
FROM cartline c
WHERE c."ORDER_REF".account_ref.customer_ref.name = 'Keanu Reeves';
```

Q2: Find the highest spending single order using MAP method)

```
SELECT
    o.account_ref.account_id AS account_id,
    o.account_ref.customer_ref.name AS cust_name,
    o.accountorder_id,
    o.total_amount AS highest_order_amount
FROM accountorder o
ORDER BY VALUE(o) DESC;
FETCH FIRST 1 ROW ONLY;
```

Q3: Compare cost of game supplier using ORDER method

```
SELECT
    sc.supplycontract_id,
    Sc.supplier_ref.name as Supplier,
    sc.cost_price,
    sc.compare_supplier(
        supplycontract_t('S011', '5/10 n/30', 45, DATE '2026-06-20',
```

```
        (SELECT REF(s) FROM supplier s WHERE s.supplier_id = 'S001')
      )
   ) as comparison
FROM supplycontract sc
WHERE sc.supplycontract_id = 'S026';
```

Q4: GenerateRecipt

```
Select c.order_ref.order_date as order_date,
        c.product_ref.name as product_name,
        c.quantity, c.order_ref.total_amount as total_amount,
        c.order_ref.account_ref.customer_ref.name as customer_name
From cartline c;
```

Q5: IsContractValid

```
SELECT sc.supplycontract_id as contract_id,
    sc.terms,
    CASE WHEN sc.expiry_date > SYSDATE THEN 'Valid' ELSE 'Expired' END as status,
    sc.expiry_date
FROM supplycontract sc
ORDER BY sc.expiry_date;
```

Q6: Subclass Override: Getprice

Videogame

```
SELECT v.product_id,
        v.name,
        v.platform,
```

```
    v.base_price AS original_price,
    v.GetPrice() AS adjusted_price,
    (v.base_price - v.GetPrice()) AS discount_amount,
    CASE WHEN v.base_price != v.GetPrice() THEN 'Discount Applied' ELSE 'No
Discount' END AS status
    FROM videogame v;
```

Accessories

```
SELECT a.product_id,
    a.name,
    a.color,
    a.base_price AS original_price,
    a.GetPrice() AS adjusted_price,
    (a.GetPrice() - a.base_price) AS premium_amount,
    CASE WHEN a.GetPrice() > a.base_price THEN 'Premium Added' ELSE 'No
Premium' END AS status
    FROM accessories a;
```

Q7: All registered customers

```
SELECT a.customer_ref.customer_id as CID,
    a.customer_ref.name as Name,
    a.customer_ref.email as Email,
    a.customer_ref.address as Address,
    a.customer_ref.phoneNum as Phone
FROM account a;
```

Q8: All customers who did not make purchase

```
SELECT c.name
FROM customer c, account a
WHERE c.customer_id = TREAT(DEREF(a.customer_ref) AS customer_t).customer_id
AND NOT EXISTS (
        SELECT 1
        FROM accountorder o
        WHERE a.account_id = TREAT(DEREF(o.account_ref) AS account_t).account_id
);
```

Q9: List supplier of all product

```
SELECT v.supplycontract_ref.supplier_ref.name as supplier_name,

v.name as product_name,

'Video Game' as product_type

FROM videogame v

UNION ALL

SELECT a.supplycontract_ref.supplier_ref.name as supplier_name,

a.name as product_name,

'Accessory' as product_type

FROM accessories a;
```

Q10: All customers who made purchases

```
SELECT DISTINCT c.name

FROM customer c, account a, accountorder o

WHERE c.customer_id = TREAT(DEREF(a.customer_ref) AS customer_t).customer_id

AND a.account_id = TREAT(DEREF(o.account_ref) AS account_t).account_id;
```

```
SQL> CREATE TYPE customer_t AS OBJECT (
  2      customer_id CHAR(5),
  3      name VARCHAR2(30),
  4      address VARCHAR2(100),
  5      email VARCHAR2(100),
  6      phoneNum CHAR(10),
  7      MEMBER PROCEDURE UpdateContactInfo(p_address VARCHAR2, p_email
VARCHAR2, p_phoneNum CHAR)
  8  );
  9  /

Type created.

SQL> CREATE TYPE account_t AS OBJECT (
  2      account_id CHAR(5),
  3      password VARCHAR2(20),
  4      guestLogin CHAR(1),
  5      customer_ref REF customer_t,
  6      MEMBER FUNCTION VerifyLogin(p_accountId VARCHAR2, p_password VARCHAR2)
RETURN CHAR,
  7      MEMBER PROCEDURE updatePassword(p_newPassword VARCHAR2)
  8  );
  9  /

Type created.

SQL> CREATE TYPE paymentmethod_t AS OBJECT (
  2      paymentmethod_id CHAR(8),
  3      payment_type CHAR(10),
  4      details VARCHAR2(60),
  5      expiry_date DATE,
  6      MEMBER FUNCTION verifyPaymentMethod RETURN CHAR
  7  );
  8  /

Type created.

SQL> CREATE TYPE supplier_t AS OBJECT (
  2      supplier_id CHAR(4),
  3      name VARCHAR2(30),
  4      address VARCHAR2(100),
  5      phone CHAR(10),
  6      rating CHAR(2),
  7      MEMBER FUNCTION GetSupplierRating RETURN CHAR
  8  );
  9  /

Type created.

SQL> CREATE TYPE supplycontract_t AS OBJECT (
  2      supplycontract_id CHAR(4),
  3      terms VARCHAR2(50),
  4      cost_price NUMBER,
  5      expiry_date DATE,
  6      supplier_ref REF supplier_t,
  7  ORDER MEMBER FUNCTION compare_supplier(other_contract IN supplycontract_t)
RETURN INTEGER
  8  );
  9  /

Type created.

SQL> CREATE TYPE product_t AS OBJECT (
  2      product_id CHAR(5),
```

```
  3      name VARCHAR2(20),
  4      base_price NUMBER,
  5      supplycontract_ref REF supplycontract_t,
  6      MEMBER FUNCTION GetPrice RETURN NUMBER,
  7      MEMBER FUNCTION SpecialPrice(p_discount NUMBER) RETURN NUMBER
  8  ) NOT FINAL;
  9  /

Type created.

SQL> CREATE TYPE videogame_t UNDER product_t (
  2      platform VARCHAR2(20),
  3      estb_rating CHAR(2),
  4      OVERRIDING MEMBER FUNCTION GetPrice RETURN NUMBER
  5  );
  6  /

Type created.

SQL> CREATE TYPE accessories_t UNDER product_t (
  2      accessory_type VARCHAR2(20),
  3      color VARCHAR2(10),
  4      OVERRIDING MEMBER FUNCTION GetPrice RETURN NUMBER
  5  );
  6  /

Type created.

SQL> CREATE TYPE accountorder_t AS OBJECT (
  2      accountorder_id CHAR(6),
  3      order_date DATE,
  4      total_amount NUMBER,
  5      account_ref REF account_t,
  6      MAP MEMBER FUNCTION get_total RETURN NUMBER
  7  );
  8  /

Type created.

SQL> CREATE TYPE cartline_t AS OBJECT (
  2      cartline_id CHAR(7),
  3      order_ref REF accountorder_t,
  4      product_ref REF product_t,
  5      quantity INTEGER,
  6      unit_price NUMBER,
  7      MEMBER FUNCTION GetLineTotal RETURN NUMBER
  8  );
  9  /

Type created.

SQL> Create table customer of customer_t (primary key (customer_id));

Table created.

SQL> Create table account of account_t (primary key (account_id));

Table created.

SQL> Create table paymentmethod of paymentmethod_t (primary key
(paymentmethod_id));

Table created.
```

```
SQL> Create table videogame of videogame_t (primary key (product_id));

Table created.

SQL> Create table accessories of accessories_t (primary key (product_id));

Table created.

SQL> Create table supplycontract of supplycontract_t (primary key
(supplycontract_id));

Table created.

SQL> Create table accountorder of accountorder_t (primary key
(accountorder_id));

Table created.

SQL> Create table cartline of cartline_t (primary key (cartline_id));

Table created.

SQL> Create table supplier of supplier_t (primary key (supplier_id));

Table created.

SQL> spool off
```

```
SQL> CREATE TYPE BODY customer_t AS MEMBER PROCEDURE UpdateContactInfo(p_address
VARCHAR2, p_email VARCHAR2, p_phoneNum CHAR) IS
  2      BEGIN
  3          IF p_address IS NOT NULL THEN
  4              self.address := p_address;
  5          END IF;
  6
  7          IF p_email IS NOT NULL THEN
  8              self.email := p_email;
  9          END IF;
 10
 11          IF p_phoneNum IS NOT NULL THEN
 12              self.phoneNum := p_phoneNum;
 13          END IF;
 14      END UpdateContactInfo;
 15  END;
 16  /

Type body created.

SQL> CREATE TYPE BODY account_t AS
  2      MEMBER FUNCTION VerifyLogin(p_accountId VARCHAR2, p_password VARCHAR2)
RETURN CHAR IS
  3      BEGIN
  4          IF self.account_id = p_accountId AND self.password = p_password
THEN
  5              RETURN 'Y';
  6          ELSE
  7              RETURN 'N';
  8          END IF;
  9      END;
 10      MEMBER PROCEDURE updatePassword(p_newPassword VARCHAR2) IS
 11      BEGIN
 12          self.password := p_newPassword;
 13          UPDATE account a
 14          SET a.password = self.password
 15          WHERE a.account_id = self.account_id;
 16      END;
 17  END;
 18  /

Type body created.

SQL> CREATE TYPE BODY paymentmethod_t AS
  2      MEMBER FUNCTION verifyPaymentMethod RETURN CHAR IS
  3      BEGIN
  4          IF self.expiry_date > SYSDATE THEN
  5              RETURN 'Y';
  6          ELSE
  7              RETURN 'N';
  8          END IF;
  9      END;
 10  END;
 11  /

Type body created.

SQL> CREATE TYPE BODY product_t AS
  2      MEMBER FUNCTION GetPrice RETURN NUMBER IS
  3      BEGIN
  4          RETURN self.base_price;
  5      END;
  6      MEMBER FUNCTION SpecialPrice(p_discount NUMBER) RETURN NUMBER IS
  7      BEGIN
```

```
  8          RETURN self.base_price * (1 - p_discount/100);
  9      END;
 10  END;
 11  /

Type body created.

SQL> CREATE TYPE BODY videogame_t AS
  2      OVERRIDING MEMBER FUNCTION GetPrice RETURN NUMBER IS
  3          v_price NUMBER;
  4      BEGIN
  5          v_price := self.base_price;
  6          IF LOWER(self.platform) = 'steam' THEN
  7              v_price := v_price * 0.9;
  8          END IF;
  9          RETURN v_price;
 10      END;
 11  END;
 12  /

Type body created.

SQL> CREATE TYPE BODY accessories_t AS
  2      OVERRIDING MEMBER FUNCTION GetPrice RETURN NUMBER IS
  3          v_price NUMBER;
  4      BEGIN
  5          v_price := self.base_price;
  6          IF LOWER(self.color) = 'gold' THEN
  7              v_price := v_price + 10;
  8          END IF;
  9          RETURN v_price;
 10      END;
 11  END;
 12  /

Type body created.

SQL> CREATE TYPE BODY supplier_t AS
  2      MEMBER FUNCTION GetSupplierRating RETURN CHAR IS
  3      BEGIN
  4          RETURN self.rating;
  5      END;
  6  END;
  7  /

Type body created.

SQL> CREATE TYPE BODY supplycontract_t AS
  2      ORDER MEMBER FUNCTION compare_supplier(other_contract IN
supplycontract_t)
  3      RETURN INTEGER IS
  4          result INTEGER := 0;
  5      BEGIN
  6          IF self.cost_price < other_contract.cost_price THEN
  7              result := -1;
  8          ELSIF self.cost_price > other_contract.cost_price THEN
  9              result := 1;
 10          END IF;
 11          RETURN result;
 12      END compare_supplier;
 13  END;
 14  /

Type body created.
```

```
SQL> CREATE TYPE BODY accountorder_t AS
  2      MAP MEMBER FUNCTION get_total RETURN NUMBER IS
  3      BEGIN
  4          RETURN self.total_amount;
  5      END;
  6  END;
  7  /

Type body created.

SQL> CREATE TYPE BODY cartline_t AS
  2      MEMBER FUNCTION GetLineTotal RETURN NUMBER IS
  3      BEGIN
  4          RETURN self.unit_price * self.quantity;
  5      END;
  6  END;
  7  /

Type body created.

SQL> spool off
```

```
SQL> Insert into customer values (customer_t('00001', 'Johnson Smith', '123
Queen St W', 'johnsons92@hotmail.com', '8475928528'));

1 row created.

SQL> Insert into customer values (customer_t('05011', 'Michael Nguyen', '940
King St W', 'michaelnguyen@gmail.com', '4378492957'));

1 row created.

SQL> Insert into customer values (customer_t('00325', 'Lucas Thompson', '55
Front St E', 'lucaslucas@gmail.com', '6472859274'));

1 row created.

SQL> Insert into customer values (customer_t('00123', 'Keanu Reeves', '156
Hollywood Dr', 'kreeves@hotmail.com', '4485163413'));

1 row created.

SQL> INSERT INTO customer VALUES (customer_t('00326', 'Sarah Mitchell', '91
Yonge St', 'sarah.mitchell@outlook.com', '4165551234') );

1 row created.

SQL> INSERT INTO account VALUES (account_t('A0001', 'password123', 'N', (SELECT
REF(c) FROM customer c WHERE c.customer_id = '00001')));

1 row created.

SQL> INSERT INTO account VALUES (account_t('A0002', 'secure456', 'N', (SELECT
REF(c) FROM customer c WHERE c.customer_id = '00123')));

1 row created.

SQL> INSERT INTO account VALUES (account_t('A0003', 'guestpass', 'Y', (SELECT
REF(c) FROM customer c WHERE c.customer_id = '00325')));

1 row created.

SQL> INSERT INTO account VALUES (account_t('A0004', 'matrix123', 'N', (SELECT
REF(c) FROM customer c WHERE c.customer_id = '05011')));

1 row created.

SQL> INSERT INTO account VALUES (account_t('A0005', 'york789', 'N', (SELECT
REF(c) FROM customer c WHERE c.customer_id = '00326')));

1 row created.

SQL> INSERT INTO paymentmethod VALUES ( paymentmethod_t( '00008462', 'e-
transfer', 'J48NSFK83HU3', LAST_DAY(TO_DATE('01/04/2026', 'DD/MM/YYYY')) ) );

1 row created.

SQL> INSERT INTO paymentmethod VALUES ( paymentmethod_t( '00084729', 'card',
'VISA', LAST_DAY(TO_DATE('30/12/2026', 'DD/MM/YYYY')) ) );

1 row created.

SQL> INSERT INTO paymentmethod VALUES ( paymentmethod_t( '00445894', 'e-
transfer', 'H94D73KDSNF8', LAST_DAY(TO_DATE('27/09/2026', 'DD/MM/YYYY')) ) );

1 row created.
```

```
SQL> INSERT INTO supplier VALUES (supplier_t('S001', 'GameDistributors Inc',
'789 Supply Ave', '4165551234', 'A'));

1 row created.

SQL> INSERT INTO supplier VALUES (supplier_t('S002', 'TechParts Ltd', '456
Industry St', '4165555678', 'B'));

1 row created.

SQL> INSERT INTO supplier VALUES (supplier_t('S003', 'Accessory World', '321
Commerce Rd', '4165559012', 'A'));

1 row created.

SQL> INSERT INTO supplier VALUES (supplier_t('S004', 'Digital Games Co', '654
Online Blvd', '4165553456', 'C'));

1 row created.

SQL> Insert into supplycontract values (supplycontract_t('S010', 'only gaming
accessories', 25, DATE '2026-03-15', (SELECT REF(s) FROM supplier s WHERE
s.supplier_id = 'S003')));

1 row created.

SQL> Insert into supplycontract values (supplycontract_t('S026', 'exclusive game
distributor', 50, DATE '2025-12-01', (SELECT REF(s) FROM supplier s WHERE
s.supplier_id = 'S001')));

1 row created.

SQL> INSERT INTO supplycontract VALUES (supplycontract_t( 'S029', 'send report
every three months', 40, DATE '2028-08-21', (SELECT REF(s) FROM supplier s WHERE
s.supplier_id = 'S004')));

1 row created.

SQL> Insert into supplycontract values (supplycontract_t('S018', 'ensure 98% on
time delivery', 15, DATE '2026-08-15', (SELECT REF(s) FROM supplier s WHERE
s.supplier_id = 'S002')));

1 row created.

SQL> INSERT INTO videogame VALUES (videogame_t('G0001', 'Cyberpunk 2077', 59.99,
(SELECT REF(s) FROM supplycontract s WHERE s.supplycontract_id = 'S026'), 'PS5',
'M'));

1 row created.

SQL> INSERT INTO videogame VALUES (videogame_t('G0002', 'Mario Kart', 49.99,
(SELECT REF(s) FROM supplycontract s WHERE s.supplycontract_id = 'S026'),
'Switch', 'E'));

1 row created.

SQL> INSERT INTO videogame VALUES (videogame_t('G0003', 'Flight Simulator',
69.99, (SELECT REF(s) FROM supplycontract s WHERE s.supplycontract_id = 'S029'),
'Steam', 'E'));

1 row created.

SQL> INSERT INTO videogame VALUES (videogame_t('G0004', 'Call of Duty', 69.99,
```

```
(SELECT REF(s) FROM supplycontract s WHERE s.supplycontract_id = 'S026'),
'XBOX', 'M'));

1 row created.

SQL> INSERT INTO accessories VALUES (accessories_t('A0001', 'HyperX Cloud',
129.99, (SELECT REF(s) FROM supplycontract s WHERE s.supplycontract_id =
'S010'), 'headset', 'black'));

1 row created.

SQL> INSERT INTO accessories VALUES (accessories_t('A0002', 'Charge Type C',
29.99, (SELECT REF(s) FROM supplycontract s WHERE s.supplycontract_id = 'S018'),
'cable', 'white'));

1 row created.

SQL> INSERT INTO accessories VALUES (accessories_t('A0003', 'PS5 Controller',
79.99, (SELECT REF(s) FROM supplycontract s WHERE s.supplycontract_id = 'S010'),
'gamepad', 'gold'));

1 row created.

SQL> INSERT INTO accessories VALUES (accessories_t('A0004', 'Logitech GPX2',
59.99, (SELECT REF(s) FROM supplycontract s WHERE s.supplycontract_id = 'S010'),
'mouse', 'silver'));

1 row created.

SQL> INSERT INTO accountorder VALUES (accountorder_t('O00001', DATE '2025-10-
15', 179.97, (SELECT REF(a) FROM account a WHERE a.account_id = 'A0001')));

1 row created.

SQL> INSERT INTO accountorder VALUES (accountorder_t('O00002', DATE '2025-10-
18', 69.99,
  2      (SELECT REF(a) FROM account a WHERE a.account_id = 'A0004')));

1 row created.

SQL> INSERT INTO accountorder VALUES (accountorder_t('O00003', DATE '2025-10-
20', 129.99, (SELECT REF(a) FROM account a WHERE a.account_id = 'A0002')));

1 row created.

SQL> INSERT INTO accountorder VALUES (accountorder_t('O00004', DATE '2025-10-
22', 189.98, (SELECT REF(a) FROM account a WHERE a.account_id = 'A0003')));

1 row created.

SQL> INSERT INTO cartline VALUES (cartline_t('CL00001',
  2      (SELECT REF(o) FROM accountorder o WHERE o.accountorder_id = 'O00001'),
  3      (SELECT REF(v) FROM videogame v WHERE v.product_id = 'G0001'),
  4      1, 59.99));

1 row created.

SQL> INSERT INTO cartline VALUES (cartline_t('CL00002',
  2      (SELECT REF(o) FROM accountorder o WHERE o.accountorder_id = 'O00001'),
  3      (SELECT REF(a) FROM accessories a WHERE a.product_id = 'A0001'),
  4      1, 129.99));

1 row created.
```

```
SQL> INSERT INTO cartline VALUES (cartline_t('CL00003',
  2      (SELECT REF(o) FROM accountorder o WHERE o.accountorder_id = 'O00002'),
  3      (SELECT REF(v) FROM videogame v WHERE v.product_id = 'G0003'),
  4      1, 69.99));

1 row created.

SQL> INSERT INTO cartline VALUES (cartline_t('CL00004',
  2      (SELECT REF(o) FROM accountorder o WHERE o.accountorder_id = 'O00003'),
  3      (SELECT REF(v) FROM videogame v WHERE v.product_id = 'G0004'),
  4      1, 69.99));

1 row created.

SQL> INSERT INTO cartline VALUES (cartline_t('CL00005',
  2      (SELECT REF(o) FROM accountorder o WHERE o.accountorder_id = 'O00004'),
  3      (SELECT REF(a) FROM accessories a WHERE a.product_id = 'A0003'),
  4      1, 79.99));

1 row created.

SQL> INSERT INTO cartline VALUES (cartline_t('CL00006',
  2      (SELECT REF(o) FROM accountorder o WHERE o.accountorder_id = 'O00004'),
  3      (SELECT REF(a) FROM accessories a WHERE a.product_id = 'A0003'),
  4      1, 79.99));

1 row created.

SQL> spool off
```

```
SQL> SELECT TREAT(DEREF(c.product_ref) AS videogame_t).platform AS platform
  2  FROM cartline c
  3  WHERE c."ORDER_REF".account_ref.customer_ref.name = 'Keanu Reeves';

PLATFORM
--------------------
XBOX

SQL> SELECT
  2      o.account_ref.account_id AS account_id,
  3      o.account_ref.customer_ref.name AS cust_name,
  4      o.accountorder_id,
  5      o.total_amount AS highest_order_amount
  6  FROM accountorder o
  7  ORDER BY VALUE(o) DESC
  8  FETCH FIRST 1 ROW ONLY;

ACCOU CUST_NAME                      ACCOUN HIGHEST_ORDER_AMOUNT
----- ------------------------------ ------ --------------------
A0003 Lucas Thompson                 O00004               189.98

SQL> SELECT
  2      sc.supplycontract_id,
  3      Sc.supplier_ref.name as Supplier,
  4      sc.cost_price,
  5      sc.compare_supplier(
  6          supplycontract_t('S011', '5/10 n/30', 45, DATE '2026-06-20',
  7              (SELECT REF(s) FROM supplier s WHERE s.supplier_id = 'S001')
  8          )
  9      ) as comparison
 10  FROM supplycontract sc
 11  WHERE sc.supplycontract_id = 'S026';

SUPP SUPPLIER                       COST_PRICE COMPARISON
---- ------------------------------ ---------- ----------
S026 GameDistributors Inc                   50          1

SQL> Select c.order_ref.order_date as order_date,
  2  c.product_ref.name as product_name,
  3  c.quantity, c.order_ref.total_amount as total_amount,
c.order_ref.account_ref.customer_ref.name as customer_name
  4  From cartline c;

ORDER_DAT PRODUCT_NAME           QUANTITY TOTAL_AMOUNT
--------- -------------------- ---------- ------------
CUSTOMER_NAME
------------------------------
15-OCT-25 Cyberpunk 2077                1       179.97
Johnson Smith

15-OCT-25 HyperX Cloud                  1       179.97
Johnson Smith

18-OCT-25 Flight Simulator              1        69.99
Michael Nguyen


ORDER_DAT PRODUCT_NAME           QUANTITY TOTAL_AMOUNT
--------- -------------------- ---------- ------------
CUSTOMER_NAME
------------------------------
20-OCT-25 Call of Duty                  1       129.99
Keanu Reeves
```

```
22-OCT-25 PS5 Controller                1       189.98
Lucas Thompson

22-OCT-25 PS5 Controller                1       189.98
Lucas Thompson


6 rows selected.

SQL> SELECT sc.supplycontract_id as contract_id,
  2         sc.terms,
  3         CASE WHEN sc.expiry_date > SYSDATE THEN 'Valid' ELSE 'Expired' END
as status,
  4         sc.expiry_date
  5  FROM supplycontract sc
  6  ORDER BY sc.expiry_date;

CONT TERMS                                               STATUS  EXPIRY_DA
---- -------------------------------------------------- ------- ---------
S026 exclusive game distributor                          Valid   01-DEC-25
S010 only gaming accessories                             Valid   15-MAR-26
S018 ensure 98% on time delivery                         Valid   15-AUG-26
S029 send report every three months                      Valid   21-AUG-28

SQL> SELECT v.product_id,
  2  v.name,
  3  v.platform,
  4  v.base_price AS original_price,
  5  v.GetPrice() AS adjusted_price,
  6  (v.base_price - v.GetPrice()) AS discount_amount,
  7  CASE WHEN v.base_price != v.GetPrice() THEN 'Discount Applied' ELSE 'No
Discount' END AS status
  8  FROM videogame v;

PRODU NAME                 PLATFORM             ORIGINAL_PRICE ADJUSTED_PRICE
----- -------------------- -------------------- -------------- --------------
DISCOUNT_AMOUNT STATUS
--------------- ----------------
G0001 Cyberpunk 2077       PS5                           59.99          59.99
              0 No Discount

G0002 Mario Kart           Switch                        49.99          49.99
              0 No Discount

G0003 Flight Simulator     Steam                         69.99         62.991
          6.999 Discount Applied


PRODU NAME                 PLATFORM             ORIGINAL_PRICE ADJUSTED_PRICE
----- -------------------- -------------------- -------------- --------------
DISCOUNT_AMOUNT STATUS
--------------- ----------------
G0004 Call of Duty         XBOX                          69.99          69.99
              0 No Discount


SQL> SELECT a.product_id,
  2  a.name,
  3  a.color,
  4  a.base_price AS original_price,
  5  a.GetPrice() AS adjusted_price,
  6  (a.GetPrice() - a.base_price) AS premium_amount,
  7  CASE WHEN a.GetPrice() > a.base_price THEN 'Premium Added' ELSE 'No
Premium' END AS status
```

```
  8  FROM accessories a;

PRODU NAME                 COLOR      ORIGINAL_PRICE ADJUSTED_PRICE
----- -------------------- ---------- -------------- --------------
PREMIUM_AMOUNT STATUS
-------------- -------------
A0001 HyperX Cloud         black              129.99         129.99
             0 No Premium

A0002 Charge Type C        white               29.99          29.99
             0 No Premium

A0003 PS5 Controller       gold                79.99          89.99
            10 Premium Added


PRODU NAME                 COLOR      ORIGINAL_PRICE ADJUSTED_PRICE
----- -------------------- ---------- -------------- --------------
PREMIUM_AMOUNT STATUS
-------------- -------------
A0004 Logitech GPX2        silver              59.99          59.99
             0 No Premium


SQL> SELECT a.customer_ref.customer_id as CID,
  2  a.customer_ref.name as Name,
  3  a.customer_ref.email as Email,
  4  a.customer_ref.address as Address,
  5  a.customer_ref.phoneNum as Phone
  6  FROM account a;

CID   NAME
----- ------------------------------
EMAIL
--------------------------------------------------------------------------------
ADDRESS
--------------------------------------------------------------------------------
PHONE
----------
00001 Johnson Smith
johnsons92@hotmail.com
123 Queen St W
8475928528


CID   NAME
----- ------------------------------
EMAIL
--------------------------------------------------------------------------------
ADDRESS
--------------------------------------------------------------------------------
PHONE
----------
00123 Keanu Reeves
kreeves@hotmail.com
156 Hollywood Dr
4485163413


CID   NAME
----- ------------------------------
EMAIL
--------------------------------------------------------------------------------
ADDRESS
```

```
--------------------------------------------------------------------------------
PHONE
----------
00325 Lucas Thompson
lucaslucas@gmail.com
55 Front St E
6472859274


CID   NAME
----- ------------------------------
EMAIL
--------------------------------------------------------------------------------
ADDRESS
--------------------------------------------------------------------------------
PHONE
----------
05011 Michael Nguyen
michaelnguyen@gmail.com
940 King St W
4378492957


CID   NAME
----- ------------------------------
EMAIL
--------------------------------------------------------------------------------
ADDRESS
--------------------------------------------------------------------------------
PHONE
----------
00326 Sarah Mitchell
sarah.mitchell@outlook.com
91 Yonge St
4165551234


SQL> SELECT c.customer_id as CID, c.name, c.email, c.phonenum
  2  FROM customer c, account a
  3  WHERE REF(c) = a.customer_ref
  4  AND NOT EXISTS (SELECT 1 FROM accountorder o WHERE REF(a) =
o.account_ref );

CID   NAME
----- ------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONENUM
----------
00326 Sarah Mitchell
sarah.mitchell@outlook.com
4165551234


SQL> SELECT v.supplycontract_ref.supplier_ref.name as supplier_name,
  2  v.name as product_name,
  3  'Video Game' as product_type
  4  FROM videogame v
  5  UNION ALL
  6  SELECT a.supplycontract_ref.supplier_ref.name as supplier_name,
  7  a.name as product_name,
  8  'Accessory' as product_type
  9  FROM accessories a;
```

```
SUPPLIER_NAME                  PRODUCT_NAME         PRODUCT_TY
------------------------------ -------------------- ----------
GameDistributors Inc           Cyberpunk 2077       Video Game
GameDistributors Inc           Mario Kart           Video Game
Digital Games Co               Flight Simulator     Video Game
GameDistributors Inc           Call of Duty         Video Game
Accessory World                HyperX Cloud         Accessory
TechParts Ltd                  Charge Type C        Accessory
Accessory World                PS5 Controller       Accessory
Accessory World                Logitech GPX2        Accessory

8 rows selected.

SQL> SELECT DISTINCT c.name
  2  FROM customer c, account a, accountorder o
  3  WHERE REF(c) = a.customer_ref AND REF(a) = o.account_ref;

NAME
------------------------------
Lucas Thompson
Keanu Reeves
Johnson Smith
Michael Nguyen

SQL> spool off
```

Video Game store System

ITEC 4220

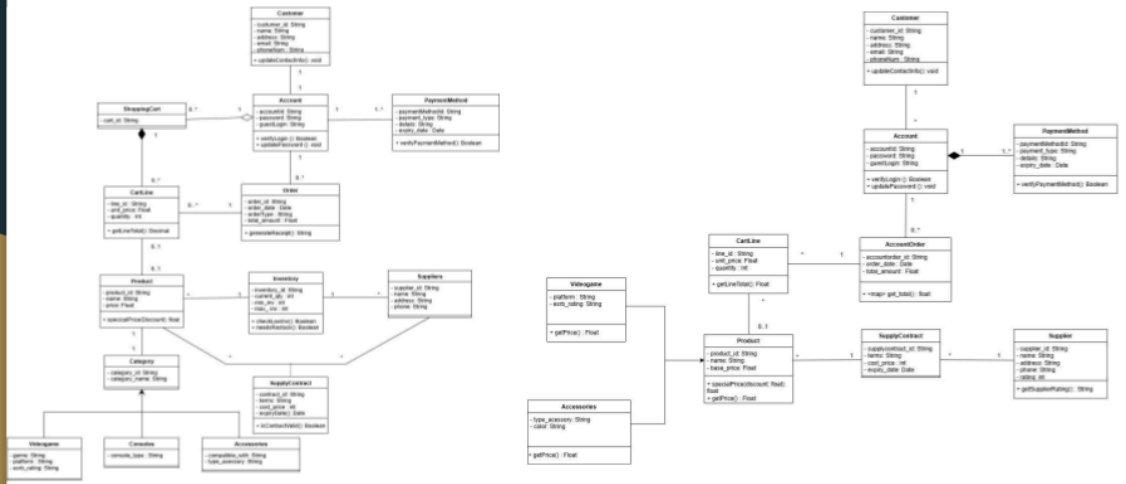
Wing Yan Kwan - 219474477

Victor Kwan - 216436990

Feng-Yuan Lee - 218501643

Brandon Lu - 218501254

Introduction:



Member methods - (Eight):

1. UpdateContact Information
2. Verifylogin
3. updatePassword
4. verifyPaymentmethod
5. GetLineTotal
6. SpecialPrice
7. GetPrice
8. GetSupplierRating

Object table:

1. Create table **customer** of customer_t (primary key (customer_id));
2. Create table **account** of account_t (primary key (account_id));
3. Create table **paymentmethod** of paymentmethod_t (primary key (paymentmethod_id));
4. Create table **videogame** of videogame_t (primary key (product_id));
5. Create table **accessories** of accessories_t (primary key (product_id));
6. Create table **supplycontract** of supplycontract_t (primary key (supplycontract_id));
7. Create table **accountorder** of accountorder_t (primary key (accountorder_id));
8. Create table **cartline** of cartline_t (primary key (cartline_id));
9. Create table **supplier** of supplier_t (primary key (supplier_id));

Type Body (Methods):

- ***Customer:***

Customer type body

```
CREATE TYPE BODY customer_t AS MEMBER PROCEDURE
UpdateContactInfo(p_address VARCHAR2, p_email VARCHAR2, p_phoneNum CHAR) IS
  BEGIN
    IF p_address IS NOT NULL THEN
      self.address := p_address;
    END IF;

    IF p_email IS NOT NULL THEN
      self.email := p_email;
    END IF;

    IF p_phoneNum IS NOT NULL THEN
      self.phoneNum := p_phoneNum;
    END IF;
  END UpdateContactInfo;
END;
/
```

- ***Account:***

Account Type Body

```
CREATE TYPE BODY account_t AS
  MEMBER FUNCTION VerifyLogin(p_accountId VARCHAR2, p_password
VARCHAR2) RETURN CHAR IS
  BEGIN
    IF self.account_id = p_accountId AND self.password = p_password THEN
      RETURN 'Y';
    ELSE
      RETURN 'N';
    END IF;
  END;
  MEMBER PROCEDURE updatePassword(p_newPassword VARCHAR2) IS
  BEGIN
    self.password := p_newPassword;
    UPDATE account a
    SET a.password = self.password
    WHERE a.account_id = self.account_id;
  END;
END;
/
```

- ***Payment method:***

Payment Method Type Body

```
CREATE TYPE BODY paymentmethod_t AS
  MEMBER FUNCTION verifyPaymentMethod RETURN CHAR IS
  BEGIN
    IF self.expiry_date > SYSDATE THEN
      RETURN 'Y';
    ELSE
      RETURN 'N';
    END IF;
  END;
END;
/
```

- ***Product:***

Product Type Body

```
CREATE TYPE BODY product_t AS
  MEMBER FUNCTION GetPrice RETURN NUMBER IS
  BEGIN
    RETURN self.base_price;
  END;
  MEMBER FUNCTION SpecialPrice(p_discount NUMBER) RETURN NUMBER IS
  BEGIN
    RETURN self.base_price * (1 - p_discount/100);
  END;
END;
/
```

- ***Video game:***

Videogame Type Body (Override Method)

```
CREATE TYPE BODY videogame_t AS
  OVERRIDING MEMBER FUNCTION GetPrice RETURN NUMBER IS
    v_price NUMBER;
  BEGIN
    v_price := self.base_price;
    IF LOWER(self.platform) = 'steam' THEN
      v_price := v_price * 0.9;
    END IF;
    RETURN v_price;
  END;
END;
/
```

- ***Accessories:***

Accessories Type Body (Override Method)

```
CREATE TYPE BODY accessories_t AS
  OVERRIDING MEMBER FUNCTION GetPrice RETURN NUMBER IS
    v_price NUMBER;
  BEGIN
    v_price := self.base_price;
    IF LOWER(self.color) = 'gold' THEN
      v_price := v_price + 10;
    END IF;
    RETURN v_price;
  END;
END;
/
```

- ***Supplier:***

Supplier Type Body

```
CREATE TYPE BODY supplier_t AS
  MEMBER FUNCTION GetSupplierRating RETURN CHAR IS
  BEGIN
    RETURN self.rating;
  END;
END;
/
```

- ***Supply Contract:***

Supply Contract Type Body with Order Method

```
CREATE TYPE BODY supplycontract_t AS
  ORDER MEMBER FUNCTION compare_supplier(other_contract IN supplycontract_t)
  RETURN INTEGER IS
    result INTEGER := 0;
  BEGIN
    IF self.cost_price < other_contract.cost_price THEN
      result := -1;
    ELSIF self.cost_price > other_contract.cost_price THEN
      result := 1;
    END IF;
    RETURN result;
  END compare_supplier;
END;
/
```

- ***Account Order:***

Account Order Type Body

```
CREATE TYPE BODY accountorder_t AS
  MAP MEMBER FUNCTION get_total RETURN NUMBER IS
  BEGIN
    RETURN self.total_amount;
  END;
END;
/
```

- ***Cartline:***

Cartline Type Body

```
CREATE TYPE BODY cartline_t AS
  MEMBER FUNCTION GetLineTotal RETURN NUMBER IS
  BEGIN
    RETURN self.unit_price * self.quantity;
  END;
END;
/
```

Data Insertion:

- ***Customer:***

Customer

```
Insert into customer values (customer_t('00001', 'Johnson Smith', '123 Queen St W',
'johnsons92@hotmail.com', '8475928528'));
Insert into customer values (customer_t('05011', 'Michael Nguyen', '940 King St W',
'michaelnguyen@gmail.com', '4378492957'));
Insert into customer values (customer_t('00325', 'Lucas Thompson', '55 Front St E',
'lucaslucas@gmail.com', '6472859274'));
Insert into customer values (customer_t('00123', 'Keanu Reeves', '156 Hollywood Dr',
'kreeves@hotmail.com', '4485163413'));
INSERT INTO customer VALUES (customer_t('00326', 'Sarah Mitchell', '91 Yonge St',
'sarah.mitchell@outlook.com', '4165551234') );
```

- ***Account with REF:***

Account with REFs

```
INSERT INTO account VALUES (account_t('A0001', 'password123', 'N', (SELECT REF(c)
FROM customer c WHERE c.customer_id = '00001')));
INSERT INTO account VALUES (account_t('A0002', 'secure456', 'N', (SELECT REF(c)
FROM customer c WHERE c.customer_id = '00123')));
INSERT INTO account VALUES (account_t('A0003', 'guestpass', 'Y', (SELECT REF(c)
FROM customer c WHERE c.customer_id = '00325')));
INSERT INTO account VALUES (account_t('A0004', 'matrix123', 'N', (SELECT REF(c)
FROM customer c WHERE c.customer_id = '05011')));
INSERT INTO account VALUES (account_t('A0005', 'york789', 'N', (SELECT REF(c)
FROM customer c WHERE c.customer_id = '00326')));
```

- ***Payment Methods:***

Payment Method

INSERT INTO paymentmethod VALUES (paymentmethod_t('00008462', 'e-transfer', 'J48NSFK83HU3', LAST_DAY(TO_DATE('01/04/2026', 'DD/MM/YYYY'))));

INSERT INTO paymentmethod VALUES (paymentmethod_t('00084729', 'card', 'VISA', LAST_DAY(TO_DATE('30/12/2026', 'DD/MM/YYYY'))));

INSERT INTO paymentmethod VALUES (paymentmethod_t('00445894', 'e-transfer', 'H94D73KDSNF8', LAST_DAY(TO_DATE('27/09/2026', 'DD/MM/YYYY'))));

- ***Suppliers:***

Suppliers

INSERT INTO supplier VALUES (supplier_t('S001', 'GameDistributors Inc', '789 Supply Ave', '4165551234', 'A'));

INSERT INTO supplier VALUES (supplier_t('S002', 'TechParts Ltd', '456 Industry St', '4165555678', 'B'));

INSERT INTO supplier VALUES (supplier_t('S003', 'Accessory World', '321 Commerce Rd', '4165559012', 'A'));

INSERT INTO supplier VALUES (supplier_t('S004', 'Digital Games Co', '654 Online Blvd', '4165553456', 'C'));

- ***Supply contract with REFs:***

Supply Contract with REFs

Insert into supplycontract values (supplycontract_t('S010', 'only gaming accessories', 25, DATE '2026-03-15', (SELECT REF(s) FROM supplier s WHERE s.supplier_id = 'S003')));

Insert into supplycontract values (supplycontract_t('S026', 'exclusive game distributor', 50, DATE '2025-12-01', (SELECT REF(s) FROM supplier s WHERE s.supplier_id = 'S001')));

INSERT INTO supplycontract VALUES (supplycontract_t('S029', 'send report every three months', 40, DATE '2028-08-21', (SELECT REF(s) FROM supplier s WHERE s.supplier_id = 'S004')));

Insert into supplycontract values (supplycontract_t('S018', 'ensure 98% on time delivery', 15, DATE '2026-08-15', (SELECT REF(s) FROM supplier s WHERE s.supplier_id = 'S002')));

- ***Videogame with REFs:***

Videogame with REFs

INSERT INTO videogame VALUES (videogame_t('G0001', 'Cyberpunk 2077', 59.99, (SELECT REF(s) FROM supplycontract s WHERE s.supplycontract_id = 'S026'), 'PS5', 'M'));

INSERT INTO videogame VALUES (videogame_t('G0002', 'Mario Kart', 49.99, (SELECT REF(s) FROM supplycontract s WHERE s.supplycontract_id = 'S026'), 'Switch', 'E'));

INSERT INTO videogame VALUES (videogame_t('G0003', 'Flight Simulator', 69.99, (SELECT REF(s) FROM supplycontract s WHERE s.supplycontract_id = 'S029'), 'Steam', 'E'));

INSERT INTO videogame VALUES (videogame_t('G0004', 'Call of Duty', 69.99, (SELECT REF(s) FROM supplycontract s WHERE s.supplycontract_id = 'S026'), 'XBOX', 'M'));

- ***Accessories with REFs:***

Accessories with REFs

```
INSERT INTO accessories VALUES (accessories_t('A0001', 'HyperX Cloud', 129.99,
(SELECT REF(s) FROM supplycontract s WHERE s.supplycontract_id = 'S010'), 'headset',
'black'));
INSERT INTO accessories VALUES (accessories_t('A0002', 'Charge Type C', 29.99,
(SELECT REF(s) FROM supplycontract s WHERE s.supplycontract_id = 'S018'), 'cable',
'white'));
INSERT INTO accessories VALUES (accessories_t('A0003', 'PS5 Controller', 79.99,
(SELECT REF(s) FROM supplycontract s WHERE s.supplycontract_id = 'S010'), 'gamepad',
'gold'));
INSERT INTO accessories VALUES (accessories_t('A0004', 'Logitech GPX2', 59.99,
(SELECT REF(s) FROM supplycontract s WHERE s.supplycontract_id = 'S010'), 'mouse',
'silver'));
```

- ***Account Orders with REFs:***

Account Orders with REFs

```
INSERT INTO accountorder VALUES (accountorder_t('O00001', DATE '2025-10-15',
179.97, (SELECT REF(a) FROM account a WHERE a.account_id = 'A0001')));
INSERT INTO accountorder VALUES (accountorder_t('O00002', DATE '2025-10-18', 69.99,
  (SELECT REF(a) FROM account a WHERE a.account_id = 'A0004')));
INSERT INTO accountorder VALUES (accountorder_t('O00003', DATE '2025-10-20',
129.99, (SELECT REF(a) FROM account a WHERE a.account_id = 'A0002')));
INSERT INTO accountorder VALUES (accountorder_t('O00004', DATE '2025-10-22',
189.98, (SELECT REF(a) FROM account a WHERE a.account_id = 'A0003')));
```

- ***CartLine with REFs:***

```
INSERT INTO cartline VALUES (cartline_t('CL00001',
  (SELECT REF(o) FROM accountorder o WHERE o.accountorder_id = 'O00001'),
  (SELECT REF(v) FROM videogame v WHERE v.product_id = 'G0001'),
  1, 59.99));
INSERT INTO cartline VALUES (cartline_t('CL00002',
  (SELECT REF(o) FROM accountorder o WHERE o.accountorder_id = 'O00001'),
  (SELECT REF(a) FROM accessories a WHERE a.product_id = 'A0001'),
  1, 129.99));
INSERT INTO cartline VALUES (cartline_t('CL00003',
  (SELECT REF(o) FROM accountorder o WHERE o.accountorder_id = 'O00002'),
  (SELECT REF(v) FROM videogame v WHERE v.product_id = 'G0003'),
  1, 69.99));
INSERT INTO cartline VALUES (cartline_t('CL00004',
  (SELECT REF(o) FROM accountorder o WHERE o.accountorder_id = 'O00003'),
  (SELECT REF(v) FROM videogame v WHERE v.product_id = 'G0004'),
  1, 69.99));
INSERT INTO cartline VALUES (cartline_t('CL00005',
  (SELECT REF(o) FROM accountorder o WHERE o.accountorder_id = 'O00004'),
  (SELECT REF(a) FROM accessories a WHERE a.product_id = 'A0003'),
  1, 79.99));
INSERT INTO cartline VALUES (cartline_t('CL00006',
  (SELECT REF(o) FROM accountorder o WHERE o.accountorder_id = 'O00004'),
  (SELECT REF(a) FROM accessories a WHERE a.product_id = 'A0003'),
  1, 79.99));
```

Query - (Ten):

1. Find the platforms of videogames ordered by customer “Keanu Reeves”
2. Find the highest spending single order using MAP method
3. Compare cost of game supplier using ORDER method
4. GenerateReceipt
5. IsContractValid
6. Subclass Override: Getpice
7. All registered customers
8. All customers who did not make purchase
9. The supplier of each product
10. All customers who made purchase

1. ***Platforms of videogames ordered by customer “Keanu Reeves” -***

```
SELECT TREAT(DEREF(c.product_ref) AS videogame_t).platform AS platform
FROM cartline c
WHERE c."ORDER_REF".account_ref.customer_ref.name = 'Keanu Reeves';
```

2. ***Find the highest spending single order using MAP method -***

```
SELECT
   o.account_ref.account_id AS account_id,
   o.account_ref.customer_ref.name AS cust_name,
   o.accountorder_id,
   o.total_amount AS highest_order_amount
FROM accountorder o
ORDER BY VALUE(o) DESC
FETCH FIRST 1 ROW ONLY;
```

3. ***Compare cost of game supplier using ORDER method -***

```
SELECT
  sc.supplycontract_id,
  Sc.supplier_ref.name as Supplier,
  sc.cost_price,
  sc.compare_supplier(
    supplycontract_t('S011', '5/10 n/30', 45, DATE '2026-06-20',
      (SELECT REF(s) FROM supplier s WHERE s.supplier_id = 'S001')
    )
  ) as comparison
FROM supplycontract sc
WHERE sc.supplycontract_id = 'S026';
```

4. ***GenerateReceipt -***

```
Select c.order_ref.order_date as order_date,
        c.product_ref.name as product_name,
        c.quantity, c.order_ref.total_amount as total_amount,
        c.order_ref.account_ref.customer_ref.name as customer_name
From cartline c;
```

5. ***IsContractValid-***

```
SELECT sc.supplycontract_id as contract_id,
    sc.terms,
    CASE WHEN sc.expiry_date > SYSDATE THEN 'Valid' ELSE 'Expired' END as status,
    sc.expiry_date
FROM supplycontract sc
ORDER BY sc.expiry_date;
```

6a. ***Subclass Override:*** **Videogame** ***Getpice -***

```
SELECT v.product_id,
    v.name,
    v.platform,
    v.base_price AS original_price,
    v.GetPrice() AS adjusted_price,
    (v.base_price - v.GetPrice()) AS discount_amount,
    CASE WHEN v.base_price != v.GetPrice() THEN 'Discount Applied' ELSE 'No
Discount' END AS status
    FROM videogame v;
```

6b. **Subclass Override: Accessories *Getpice* -**

```
SQL> SELECT a.product_id,
  2  a.name,
  3  a.color,
  4  a.base_price AS original_price,
  5  a.GetPrice() AS adjusted_price,
  6  (a.GetPrice() - a.base_price) AS premium_amount,
  7  CASE WHEN a.GetPrice() > a.base_price THEN 'Premium Added' ELSE 'No
Premium' END AS status
  8  FROM accessories a;
```

7. ***All registered customers -***

```
SELECT a.customer_ref.customer_id as CID,
a.customer_ref.name as Name,
a.customer_ref.email as Email,
a.customer_ref.address as Address,
a.customer_ref.phoneNum as Phone
FROM account a;
```

8. ***All customers who did not make purchase -***

```
SELECT c.customer_id as CID, c.name, c.email, c.phonenum
FROM customer c, account a
WHERE REF(c) = a.customer_ref
AND NOT EXISTS (SELECT 1 FROM accountorder o WHERE REF(a) = o.account_ref );
```

9. ***The supplier of each product -***

```
SELECT v.supplycontract_ref.supplier_ref.name as supplier_name,
v.name as product_name,
'Video Game' as product_type
FROM videogame v
UNION ALL
SELECT a.supplycontract_ref.supplier_ref.name as supplier_name,
a.name as product_name,
'Accessory' as product_type
FROM accessories a;
```

10. ***All customers who made purchase -***

```
SELECT DISTINCT c.name
FROM customer c, account a, accountorder o
WHERE REF(c) = a.customer_ref AND REF(a) = o.account_ref;
```

Conclusion

